Practice Exam

If any information is not specified, it is up to you to decide!

Task I. (4 points)

Create a client-server application that uses the TCP protocol.
The server represents a lab instructor from the Telecommunications Networks course.
The clients are the students attending the lab who want to receive a practice exam.

The client must send the message **"I would like a task"** to the server.
The server initially replies **"Not yet"**.
When a sufficient number of students (between 1–5) have requested a task, the instructor finally decides to create one and replies: **"Here is the task!"**.
After the students receive the task, they thank the instructor with a **"Thank you"** message, to which the instructor responds with **"You're welcome"**.
All messages must be of type **bytes**!

Task II. (1 point)

The required (sufficient) number of students must be a parameter of the server (an integer between 1 and 5 inclusive)!

Task III. (1 point)

The instructor's server must be able to serve multiple students concurrently.
This must be implemented using the **select** function!

Task IV. (4 points)

Create a second server as well: the "Previous Years' Exam Server".
When the instructor decides to create the task, he/she checks the previous years' exam tasks — that is, the instructor server sends a **"Search"** message to the Previous Years' Exam Server (via UDP).
The Previous Years' Exam Server responds with a message **"taskX"**, where X is a random number between 1 and 10.
Only after receiving this does the instructor forward the task to the students.
Communication between the instructor server and the Previous Years' Exam Server must use the **UDP protocol**.